TORONTO NEUROIMAGING

# The Importance and Benefits of Open Science

## *Practical Lessons for Sustainable MR Research*

*Open Science for MR in RT,*

*MR in Radiation Therapy Study Group Virtual Meeting*

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*With many thanks for slides and input:*

*Johanna Bayer*

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*Slides available on: mrikasper.github.io*

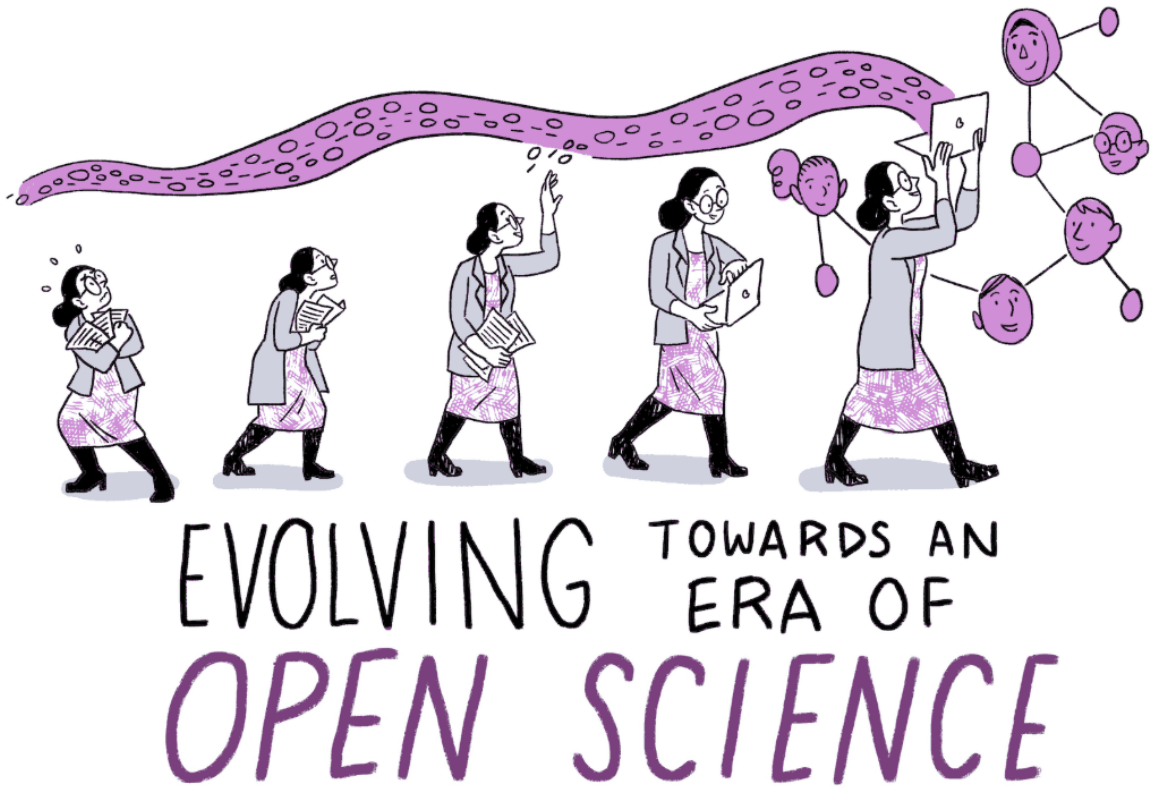
# Declaration of Financial Interests or Relationships

Speaker Name: Lars Kasper, PhD

I have no financial interests or relationships to disclose with regard to the subject matter of this presentation.

# Why I care about this …

- MR Physicist — from reconstruction to clinical translation (psychiatry)
- Built tools used beyond my own lab (TAPAS, PhysIO, UniQC, GIRFReco.jl)
- Learned the hard way about legacy code, lost detail, and sustainability
- Realized open science is not idealism, but *effective infrastructure*
  - for yourself and others



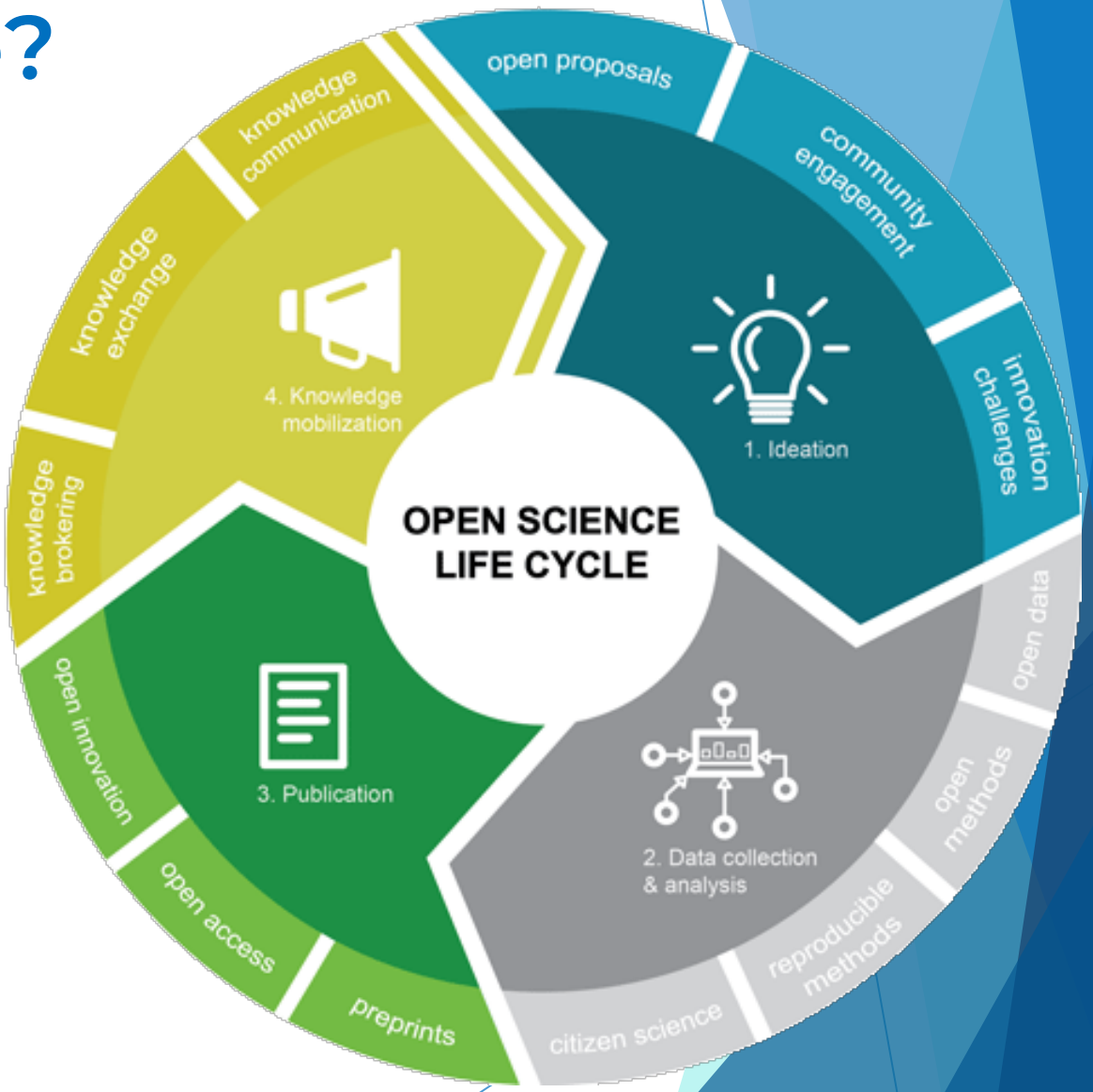

*More info (incl. these slides): [mrikasper.github.io](mrikasper.github.io)*

*The Turing Way project illustration by Scriberia. Used under a CC-BY 4.0 licence. DOI: [The Turing Way Community & Scriberia (2024)](#).*

# What is Open Science?

- Open science means structuring the **entire research lifecycle** for **transparency** and **reuse**

1. *Ideation:* Registered reports
2. *Data Collection & Analysis:* Data Sharing, Open Source Code, Open Hardware
3. *Publication:* Open Access, standardized reporting (protocols), preprints, Open & Code Review
4. *Knowledge Mobilization:* Communities, Challenges, Reuse/Reproduce




*Gaspar et al. (2025), "Advancing MRI, together: open science in MR research" (MAGMA, doi:10.1007/s10334-025-01286-8)*
https://science.gc.ca/site/science/en/open-science

# Example (Step 2): FAIR Data & Code

## Findable

The first step in (re)using data/code is to find it!

Descriptive metadata (title, authors, keywords, version) is essential

Code should be deposited in a repository with a persistent identifier (e.g., DOI)

## Accessible

Once the user finds data and software they need to know how to access it.

Data could be openly available;

Also possible that authentication and authorisation is necessary.

The repository should be publicly available or provide clear access procedures, version control, and instructions for installation and dependencies.

## Interoperable

Data and code need to be interoperable for humans and machines!

Data needs to be integrated with other data and interoperate with applications or workflows.

Follow community standards and use interoperable formats

Libraries, and APIs integrated with other software, workflows, or data. Clear documentation of inputs and outputs supports interoperability.

## Reusable

Share all information about your data, code and its provenance.

Data should be well-described so that they can be used, combined, and extended in different settings.

Code should be well-documented, versioned, and released under an explicit open-source license.

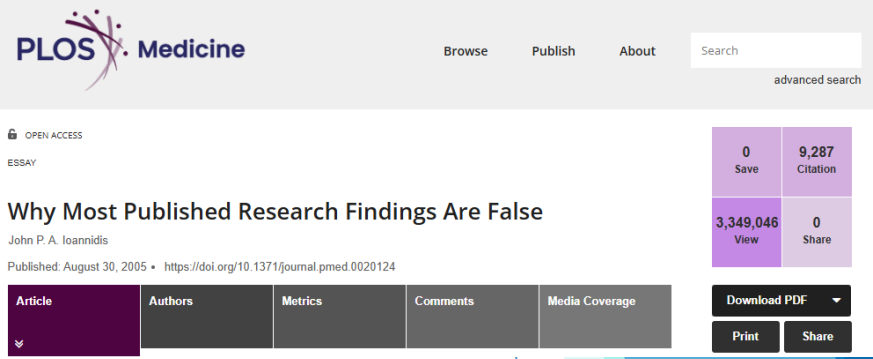
Include examples or tests so that others can understand, adapt, and extend it for new contexts.

https://www.openaire.eu/how-to-make-your-data-fair

*Slide courtesy of Johanna Bayer*

# Why Open Science?

- There are strong systemic and ethical arguments for open science,
  - Improving reproducibility
  - Reducing false positives[1]
- This sounds like a lot of work…what's in it for me?
- Today: Practical (and slightly selfish) arguments[2]
  1. Efficiency & Error Reduction
  2. Collaboration & Continuity
  3. Impact & Influence



*https://lccn.loc.gov/50041709, Public domain, via Wikimedia Commons*

[1] Ioannidis, J.P.A., 2005. Why Most Published Research Findings Are False. PLOS Medicine 2, e124. https://doi.org/10.1371/journal.pmed.0020124
*[2] Markowetz, F., 2015. Five selfish reasons to work reproducibly. Genome Biol 16, 274. https://doi.org/10.1186/s13059-015-0850-7*

# Practical Lesson 1: Reduce Friction -
## *Efficiency & Error Reduction*

- Saves time when revisiting analyses
- Prevents silent errors before submission
- Makes debugging and parameter tracking feasible
- Makes writing and revising papers easier
- Strengthens your position during peer review

- *"I did this analysis 6 months ago. Of course I can't remember all the details after such a long time"*[1]
- *E.g., Code Review by other co-author*[2]
- *Method section writes itself*
- *"Rev 2 wants to try a different setting for a processing step"*

[1] Markowetz, F., 2015. Five selfish reasons to work reproducibly. Genome Biol 16, 274. https://doi.org/10.1186/s13059-015-0850-7
[2] Pereira, I., et al., (2025). Structured Code Review in Translational Neuromodeling and Computational Psychiatry. https://doi.org/10.20944/preprints202510.0133.v1

# Practical Lesson 2: Enable Sustainability

## *Collaboration & Continuity*

- Faster onboarding of students and staff
- Pipelines survive personnel changes
- Multi-center validation becomes feasible
- Infrastructure outlives individual projects
- Reduces technical debt

- *What's documented for the public, is well documented for your own lab/center*[1]

*Research finds organizations that contribute to open source report financial and operational gains, while private forks and workarounds drain labor hours and budgets*

- *Proprietary software or private branches of open source software: licensing or internal software engineering costs*[2]

[1] Frässle, S. et al. (2021) TAPAS: An Open-Source Software Package for Translational Neuromodeling and Computational Psychiatry. Front. Psychiatry 12. https://doi.org/10.3389/fpsyt.2021.680811
[2] https://www.linuxfoundation.org/press/new-linux-foundation-report-shows-active-open-source-contribution-delivers-2-5x-roi-while-passive-consumption-increases-costly-technical-debt

# Practical Lesson 3: Increase Leverage

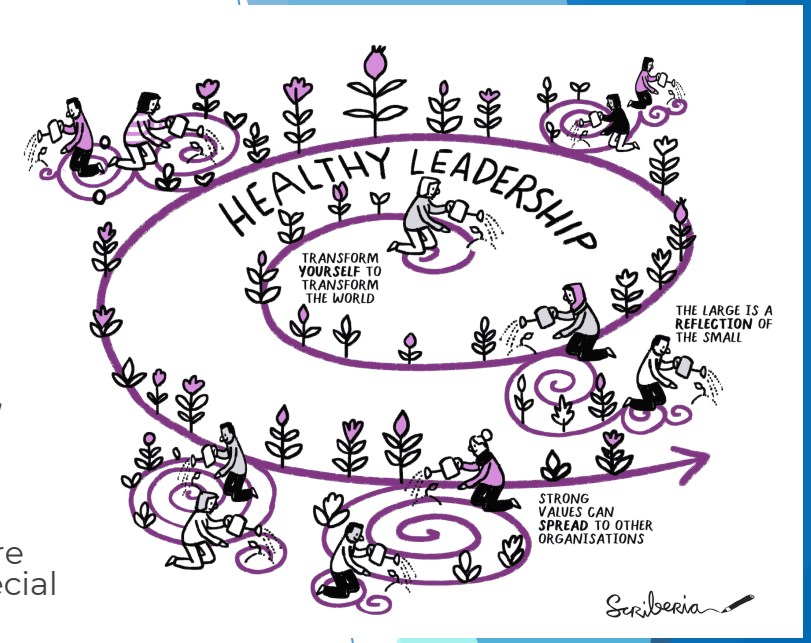
## *Impact & Influence*

- Shared data and code are reused — and reused work is cited more
- Methods become benchmarks, not just papers
- Stronger position in multi-center collaborations
- Alignment with evolving journal and funder requirements
- Reputation for rigor and transparency

- *Up to 25% higher citation rates[1]*
- *E.g., conjugate gradient SENSE for image reconstruction challenge of ISMRM RRSG[2]*
- *Code and Data Sharing increasingly recommended or even required by journals[3]*

*[1] Colavizza, G., et al. (2020). The citation advantage of linking publications to research data. PLOS ONE 15, e0230416.* https://doi.org/10.1371/journal.pone.0230416
*[2] Maier, O.,et al (2021). CG-SENSE revisited: Results from the first ISMRM reproducibility challenge. MRM 85, 1821–1839.* https://doi.org/10.1002/mrm.28569
*[3] Pariente, N., Cadwallader, L., Editors, on behalf of the P.B.S., 2026. Formalizing our commitment to code sharing. PLOS Biology 24, e3003678.* https://doi.org/10.1371/journal.pbio.3003678

# The lids on unopened science – What is holding us back?




- Resource scarcity: Time and staff
- Activation energy:
  - Where do I start?
  - Perfectionism
- Templates, AI coding tools (e.g., Claude, Codex, Copilot, Cursor)
- Seek out communities
  - Education: Turing Way, Software Carpentry, Domain-specific special issues/review papers
  - One commit(ment) at a time
- Institutional/domain incentive structure
  - Lab culture
  - Career progression
  - Prestige
- Legal Concerns
  - Personal Health Information,
  - Data Sharing/GDPR
  - Intellectual Property/Licensing
- Advocacy & Communities:
  - PIs: Collaborative Group culture
  - Publishers/Editors
  - Study/Special Interest Groups of Academic Societies
  - Funders/Paper/Grant Reviewers (Data Availability statements.)

*The Turing Way Community & Scriberia (2024)*
*https://doi.org/10.5281/ZENODO.3332807*

# Getting started with Open Science Today

- Find a great community
  - Open and Reproducible Research Study Group (ISMRM)
  - Open Source Hardware Initiative (https://opensourceimaging.org)
  - Open Science Initiative for Perfusion Imaging (https://osipi.ismrm.org/)
  - Glioma MR Imaging 2.0 (GliMR) Network[1]
    - Ultimate consent form - GDPR edition — Open Brain Consent 1.1.0+some documentation
  - Open and Reproducible MSK Imaging Research Community (https://www.ormir.org/)
  - MRITogether (ESMRMB)
  - The Turing Way[2], Slogan: The Turing Way – Make Open Science *"too easy NOT to do" it*

- Nice special issues/primers to start reading:
  - *Gaspar, A et al. (2025). Advancing MRI, together: open science in MR research. MAGMA 38, 635–638. https://doi.org/10.1007/s10334-025-01286-8*
  - *Tamir, J.I., et al. (2025. MRI acquisition and reconstruction cookbook: recipes for reproducibility, served with real-world flavour (MAGMA 38), 367–385. https://doi.org/10.1007/s10334-025-01236-4*

*[1] Clement, P., et al. (2021). GliMR: Cross-Border Collaborations to Promote Advanced MRI Biomarkers for Glioma. J. Med. Biol. Eng. 41, 115–125. https://doi.org/10.1007/s40846-020-00582-z*
*[2] The Turing Way Community & Scriberia (2024) https://doi.org/10.5281/ZENODO.3332807*

# Resources I


- Clement, P., Booth, T., Borovečki, F., Emblem, K.E., Figueiredo, P., Hirschler, L., Jančálek, R., Keil, V.C., Maumet, C., Özsunar, Y., Pernet, C., Petr, J., Pinto, J., Smits, M., Warnert, E.A.H., 2021. GliMR: Cross-Border Collaborations to Promote Advanced MRI Biomarkers for Glioma. J. Med. Biol. Eng. 41, 115–125. https://doi.org/10.1007/s40846-020-00582-z
- Colavizza, G., Hrynaszkiewicz, I., Staden, I., Whitaker, K., McGillivray, B., 2020. The citation advantage of linking publications to research data. PLOS ONE 15, e0230416. https://doi.org/10.1371/journal.pone.0230416
- Community, T.T.W., 2025. The Turing Way: A handbook for reproducible, ethical and collaborative research. https://doi.org/10.5281/zenodo.15213042
- Frässle, S., Aponte, E.A., Bollmann, S., Brodersen, K.H., Do, C.T., Harrison, O.K., Harrison, S.J., Heinzle, J., Iglesias, S., Kasper, L., Lomakina, E.I., Mathys, C., Müller-Schrader, M., Pereira, I., Petzschner, F.H., Raman, S., Schöbi, D., Toussaint, B., Weber, L.A., Yao, Y., Stephan, K.E., 2021. TAPAS: An Open-Source Software Package for Translational Neuromodeling and Computational Psychiatry. Front. Psychiatry 12. https://doi.org/10.3389/fpsyt.2021.680811
- Gaspar, A., Nagtegaal, M., Santini, F., Schauman, S., Shahdloo, M., Van Houdt, P.J., Wang, Y.-F., Webb, A., 2025. Advancing MRI, together: open science in MR research. Magn Reson Mater Phy 38, 635–638. https://doi.org/10.1007/s10334-025-01286-8
- Ioannidis, J.P.A., 2005. Why Most Published Research Findings Are False. PLOS Medicine 2, e124. https://doi.org/10.1371/journal.pmed.0020124
- Maier, O., Baete, S.H., Fyrdahl, A., Hammernik, K., Harrevelt, S., Kasper, L., Karakuzu, A., Loecher, M., Patzig, F., Tian, Y., Wang, K., Gallichan, D., Uecker, M., Knoll, F., 2021. CG-SENSE revisited: Results from the first ISMRM reproducibility challenge. Magnetic Resonance in Medicine 85, 1821–1839. https://doi.org/10.1002/mrm.28569
- Markowetz, F., 2015. Five selfish reasons to work reproducibly. Genome Biol 16, 274. https://doi.org/10.1186/s13059-015-0850-7
- Masum, H., Rao, A., Good, B.M., Todd, M.H., Edwards, A.M., Chan, L., Bunin, B.A., Su, A.I., Thomas, Z., Bourne, P.E., 2013. Ten Simple Rules for Cultivating Open Science and Collaborative R&D. PLOS Computational Biology 9, e1003244. https://doi.org/10.1371/journal.pcbi.1003244
- Maumet, C., n.d. Open science practices for keeping inventions alive after project ends. Statistical analysis. https://inria.hal.science/hal-04236827v1/file/Maumet_ESMRMB_keeping%20inventions%20alive%20after%20project%20ends.pdf
- New Linux Foundation Report Shows Active Open Source Contribution Delivers 2-5x ROI, While Passive Consumption Increases Costly Technical Debt [WWW Document], n.d. URL https://www.linuxfoundation.org/press/new-linux-foundation-report-shows-active-open-source-contribution-delivers-2-5x-roi-while-passive-consumption-increases-costly-technical-debt (accessed 3.2.26).
- Pariente, N., Cadwallader, L., Editors, on behalf of the P.B.S., 2026. Formalizing our commitment to code sharing. PLOS Biology 24, e3003678. https://doi.org/10.1371/journal.pbio.3003678
- Pereira, I., Galioulline, H., Enz, R., Hess, A., Müller-Schrader, M., Von Werder, D., Kertesz, I., Schönleitner, F., Brand, K., Kasper, L., Weber, L., Petzschner, F., Iglesias, S., Heinzle, J., Stephan, K., 2025a. Structured Code Review in Translational Neuromodeling and Computational Psychiatry. https://doi.org/10.20944/preprints202510.0133.v1
- Pereira, I., Galioulline, H., Enz, R., Hess, A., Müller-Schrader, M., Von Werder, D., Kertesz, I., Schönleitner, F., Mellor, S., Brand, K., Kasper, L., Weber, L., Petzschner, F., Iglesias, S., Heinzle, J., Stephan, K., 2025b. Structured Code Review in Mental Health Research. https://doi.org/10.20944/preprints202510.0133.v2
- Reproducibility and Quality Assurance [WWW Document], n.d. . SpringerLink. URL https://link.springer.com/collections/ehhdhbedab (accessed 2.27.26).
- Tamir, J.I., Blumenthal, M., Wang, J., Oved, T., Shimron, E., Zaiss, M., 2025. MRI acquisition and reconstruction cookbook: recipes for reproducibility, served with real-world flavour. Magn Reson Mater Phy 38, 367–385. https://doi.org/10.1007/s10334-025-01236-4
- The Turing Way [WWW Document], n.d. URL https://book.the-turing-way.org/ (accessed 3.2.26).
- Zuckerman, A.L., Faber, S., Shen, K., McIntosh, A.R., Juavinett, A.L., 2026. Ten simple rules for building a collaborative coding culture. PLOS Computational Biology 22, e1013970. https://doi.org/10.1371/journal.pcbi.1013970

# Resources II – Easy Listening

- MRI Together 2021 - Opening (Atlantic) - Introduction to Open Science… (C. Gould van Praag)
  - *A much more in-depth introduction on open science and its development*
- MRITogether2024-S7T1: Open Science and Data Sharing using the CAMH BrainHealth Databank (Nicole Schoer)
  - *Data Sharing – Legal Concerns; Example for North America (mental health)*
- MRI Together 2023 - Session 6 Talk 4 - PhysIO, UniQC and a TAPAStry of Tools… (L. Kasper)
  - *An intro into open-source tools and their development philopsophy I am heavily involved in*

# Thank you for your attention!

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